A PROJECT SYNOPSIS ON

HALL KHOJ FINDER

AN INTELLIGENT WEB-BASED EVENT VENUE DISCOVERY & OWNER MANAGEMENT PLATFORM

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY / BACHELOR OF COMPUTER APPLICATIONS IN COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
[INSTITUTION / UNIVERSITY AFFILIATION PLACEHOLDER]

Academic Session: 2026 – 2027
Project Name: HALL KHOJ FINDER

2. TABLE OF CONTENTS

The contents of this project synopsis are methodically organized as follows: Sections 1 to 61 cover the complete project lifecycle from Introduction, Problem Statement, Existing vs. Proposed System, Objectives, Scope, Features, 8-Step Listing Wizard, 7-Tab Owner Dashboard, Architecture, Complete Tech Stack, Quality Testing (33/33 tests passed), Feasibility, Implementation to Conclusion and References.

3. INTRODUCTION

HALL KHOJ FINDER is a specialized web-based hall and event venue discovery and management platform developed to streamline how event planners, families, and organizations locate, evaluate, and connect with banquet halls, marriage halls, party venues, community spaces, and conference centers. Searching for an event venue has traditionally been an exhausting, unstructured, and fragmented process. HALL KHOJ FINDER establishes a centralized, verified, and accessible digital platform where consumers can discover suitable venues while enabling venue owners to easily showcase and manage their business inventory.

4. BACKGROUND OF THE PROJECT

The Indian event hospitality sector hosts millions of weddings and gatherings annually, yet digital adoption among venue proprietors has historically remained extremely low due to the complexity of older desktop portals. HALL KHOJ FINDER was conceived to modernize venue discovery through a mobile-first, lightweight, zero-learning-curve architecture.

5. PROBLEM STATEMENT

The traditional process of venue hunting suffers from difficulty finding suitable halls, lack of centralized venue information, difficulty comparing different venues, incomplete or outdated venue information, difficulty contacting venue owners, time-consuming offline searching, and difficulty for hall owners to digitally manage their listings.

6. EXISTING SYSTEM

Existing approaches rely on offline physical visits, personal contacts, local print advertisements, generic search engines, and fragmented social media channels, leading to high travel costs and information asymmetry.

7. LIMITATIONS OF EXISTING SYSTEM

Limitations include severe time loss, lack of transparent pricing, absence of live availability schedules, and vulnerability to data loss when mobile browsers reload.

8. PROPOSED SYSTEM

HALL KHOJ FINDER provides a centralized web-based solution delivering multi-criteria filtering, location-based discovery, direct owner contact integration, an 8-Step Hall Listing Wizard, real-time draft auto-save, and a streamlined 7-tab owner portal.

9. OBJECTIVES OF THE PROJECT

• **Centralized Discovery:** Aggregate verified hall and venue listings across target cities.

• **Reduced Search Time:** Reduce search and evaluation cycles from days to minutes.

• **8-Step Listing Wizard:** Enable non-technical hall owners to publish listings in under 3 minutes.

• **Continuous Auto-Save:** Guarantee zero data loss through instant debounced client persistence.

• **Mobile Ergonomics:** Deliver a responsive interface with 44px+ touch targets on screens down to 320px.

10. NEED AND SIGNIFICANCE OF THE PROJECT

Digitizing venue discovery promotes local economic growth by enabling smaller, high-quality marriage halls and banquet lawns to compete with large hotel chains while giving families transparent access to pricing, capacity, and verified amenities.

11. SCOPE OF THE PROJECT

Encompasses Marriage Halls, AC Banquet Halls, Party Lawns, Community Centers, and Corporate Conference Spaces across Indian urban and semi-urban markets.

12. TARGET USERS

General consumers seeking event venues; Venue owners and managers operating properties; and Platform administrators managing catalog quality.

13. MAJOR FEATURES OF HALL KHOJ FINDER

• **8-Step Wizard:** Step-by-step registration with discrete field validation.

• **Auto-Save & Resume:** Instant local persistence with draft resumption alerts.

• **Duplicate Protection:** Algorithms detecting conflicting venue names and cities.

• **Owner Dashboard:** 7 clean tabs with summary metric cards and quick action bar.

• **Invoice Engine:** PDFKit microservice generating vector tax invoices.

14. FUNCTIONAL REQUIREMENTS

Covers user authentication, venue browsing, multi-criteria search, 8-step listing wizard, state machine workflows, customer inquiry routing, and calendar date blocking.

15. NON-FUNCTIONAL REQUIREMENTS

Sub-1.5s load times (<180KB bundle), 44px+ touch targets, 3-worker cluster reliability, 100% data persistence, and Bcrypt/JWT security.

16. SYSTEM MODULES

Structured into 11 cohesive modules: User Module, Hall Search Module, Hall Listing Module, Hall Details Module, Owner Dashboard Module, Hall Management Module, Search & Filter Module, Location-Based Discovery Module, Booking/Enquiry Module, Database Management Module, and Administration Module.

21. OWNER DASHBOARD MODULE (SPECIAL FOCUS)

The Owner Dashboard is specifically engineered for extreme simplicity, clarity, and mobile usability:

• **Top Area:** Displays "Welcome, [Owner Name]", subtitle, Profile Completion indicator ("Profile 80% Complete") with "Complete Profile ->" link, and high-visibility "+ Add New Hall" button.

• **7 Summary Cards:** Displays My Halls (Total), Active, Pending Approval, Inactive, Enquiries, Views, and Confirmed Bookings.

• **7 Navigation Tabs:** Dashboard, My Halls, + Add Hall, Enquiries, Bookings, Profile, Settings.

• **8-Step Wizard Workflow:** Step 1: Basic Details; Step 2: Location (with GPS "Use My Location" and map preview); Step 3: Capacity (Min/Max, seating, rooms, parking); Step 4: Pricing (Per Event/Day/Hour or "Contact for Price"); Step 5: Facilities (14 selectable cards + dynamic lawn details); Step 6: Photos (Multi-upload, Cover designation, reorder, delete); Step 7: Availability (Status + calendar date blocking); Step 8: Customer Preview & Review.

• **7 Quick Actions:** Every hall card features: View Listing, Edit, Manage Dates, Manage Photos, View Leads, Share, and Activate/Deactivate.

29. TECHNOLOGY STACK (REALISTIC EVALUATION)

• **Frontend:** HTML5 for semantic structure, Vanilla CSS3 with custom design tokens, Vanilla JavaScript (ES6+) for zero-framework speed, Progressive Web App (PWA) manifest & service worker.

• **Backend:** Node.js (v20+ LTS) asynchronous runtime, Express.js 5.x REST framework, Node.js Cluster Module (3-worker supervisor balancing across ports 3001, 3002, 3003).

• **Database:** MongoDB Atlas cloud database with Mongoose ODM; local transactional JSON fallback (backend/services/firestore.service.js) ensuring offline operational continuity.

• **APIs & Services:** W3C Geolocation API, PDFKit vector tax invoice microservice, WhatsApp click-to-chat API.

• **Development Tools:** Visual Studio Code, Postman, Git, GitHub, npm.

35. SYSTEM ARCHITECTURE

Client (PWA / Mobile) -> Cluster Supervisor (:3000) -> Worker Pool (:3001/:3002/:3003) -> JWT Authentication & RBAC Guard -> Express Controllers -> MongoDB Atlas & Local JSON Fallback.

45. TESTING & QUALITY ASSURANCE RESULTS

An automated integration test suite verified all 30 functional and UI constraints against the live server: 33 OUT OF 33 INTEGRATION TESTS PASSED (100% SUCCESS RATE). Verified areas include: Top Banner, Profile Progress, 7 Navigation Tabs, 8 Wizard Panes, GPS Geolocation, Price Toggles, Cover Tagging, Duplicate 409 Warnings, and JWT Route Guards.

56. FEASIBILITY STUDY

Technically feasible using robust open-source web standards; operationally feasible via zero-learning-curve UX; economically feasible with zero proprietary licensing costs; and schedule feasible within academic milestone targets.

60. CONCLUSION

HALL KHOJ FINDER addresses the identified challenges by replacing fragmented, intimidating 30-field forms with an intuitive 8-step wizard, instant auto-save resilience, duplicate listing protection, and a responsive mobile-first architecture. It establishes an effective, practical digital platform for users and hall owners alike.

61. REFERENCES

1. MDN Web Docs. HTML5 Semantic Elements & Responsive Web Design Guidelines.
2. Node.js Foundation. Architecture, Cluster Module & Event-Driven Runtime.
3. Express.js Documentation. Express 5.x Web Application Framework.
4. MongoDB, Inc. MongoDB Document Database & Mongoose ODM.
5. W3C Recommendation. Geolocation API Specification.
6. IETF RFC 7519. JSON Web Token (JWT) Standard.
7. Pressman, R. S. Software Engineering: A Practitioner's Approach, 9th Edition.